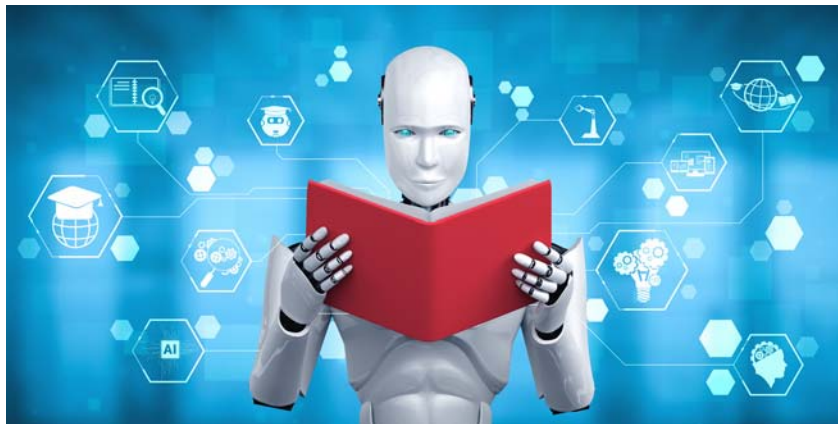


A Brief History and the Benefits of Shogun, the ML Library for Everyone



Shogun is an open source machine learning software library built in C++. It offers a wide range of efficient and unified machine learning algorithms. The heart of Shogun lies in kernel machines such as support vector machines for regression and classification problems. Read on to understand more about this library.

Shogun offers a full implementation of Hidden Markov models. Its core is written in C++ and it offers interfaces for MATLAB, Octave, Python, R, Java, Lua, Ruby and C#. Today, there is a large and active user community all over the world that uses Shogun as a base for education and research, and also contributes to the core package.

Shogun was developed in 1999 by Soeren Sonnenburg and Gunnar Raetsch. The inventive focus was on large-scale kernel methods and bioinformatics. Since then, it has continuously been used for scientific research and has expanded massively. In early 2017, Shogun joined NumFocus.

Presently, Shogun is developed by a diverse team of students, scientists and

professionals. None of their work would have been possible, however, without the patches and bug reports given by various contributors.

Mission

Machine learning is powerful. Since the last two decades, machine learning (ML) has emerged from an unconventional academic field to changing the way we live. Its methods impact society and its tools have a billion-dollar influence on the global economy. Shogun developers want to make sure that the primary algorithms are user-friendly and accessible. They want users like scientists, data enthusiasts, journalists, students, non-profit organisations and companies, as well as common citizens to be able to use ML tools at no cost.

The Shogun ML toolkit encourages this expedition through its features (accessible, open source and good, which will be explained later) and a pivot on ML education and development. Shogun is one of the largest and oldest open source ML platforms.

Features

Given below are some features of Shogun which make it unique.

Accessible: Shogun is not limited to any single language; you can use the toolkit through a unified interface (SWIG) from C++, Python, Octave, R, Java, Lua, C#, Ruby, etc. Thus Shogun is independent of trends in computing languages and importantly, this allows us to use it as a vehicle to expose one's algorithms to multiple communities. Shogun runs natively under Linux/UNIX, MacOS and Windows.

Anybody can attempt and learn Shogun with minimal effort, simply by connecting to a Jupyter Notebook (an open source Web application that allows us to create and share documents that contain live code, equations or other things). Users can run Shogun in the cloud. It also offers this service for free through external donations.

Shogun does not reinvent anything, but provides bindings to other sophisticated libraries like Tapkee, SLEP, GPML, LibSVM/LibLinear, SVMlight, LibOCAS, libqp, VowpalWabbit, and more.

Free (open source): Shogun is community-based and non-commercial. It is currently released under the GPLv3. It is also moving towards BSD compatibility with optional GPL (General Public License) parts. This